COOPER MCALLISTER

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Seeking a summer engineering internship in cutting-edge electronics and controls technology. Hands-on experience working in design, development, and testing of multidisciplinary projects, both on teams and individually.

EDUCATION

LETOURNEAU UNIVERSITY, ELECTRICAL AND COMPUTER ENGINEERING, MECHATRONICS CONCENTRATION, B.S. CLASS OF 2027 | Honors College | President's List | GPA 4.0

WORK EXPERIENCE

AVIONICS AND ELECTRICAL SYSTEMS INTERN, TEXTRON AVIATION

JUNE 2025 – AUGUST 2025

- Wrote Python scripts to automate Garmin G1000 avionics testing for Cessna aircraft, decreasing testing time by 80%
- Gained experience with Siemens Capital (Mentor Graphics), MS Office, VS Code, and Textron software tools
- Assisted with design changes due to part obsolescence and created documentation for production functional tests

AVIONICS TEAM LEAD, LETOURNEAU UNIVERSITY SPACE PROGRAM

AUGUST 2025 – PRESENT (LEAD), AUGUST 2024 – AUGUST 2025 (MEMBER)

- Lead a team of six to manage development of avionics systems with live telemetry for high-power amateur rocketry
- PCB and software integration using KiCad, VS Code, GitHub, Raspberry Pi, and serial peripherals programmed in C
- Collaborate with rocketry and propulsion teams to integrate avionics objectives, such as adding cameras and sensors

ELECTRIC CIRCUITS AND MATH TUTOR, LETOURNEAU UNIVERSITY

JANUARY 2025 – PRESENT

- Peer tutor 2-3 hours a week for Electric Circuits, Calculus, and Differential Equations
- Provide explanations and example problems to help increase students' understanding of core concepts

PROJECT EXPERIENCE

PCB DESIGN, EMBEDDED SYSTEMS, CAD DESIGN, PROJECT MANAGEMENT - Rocket Motor Advanced Remote Ignition And Evaluation System (ARIES) – In Progress

AUGUST 2025 – PRESENT

- Responsible for circuit design, PCB design, CAD design, soldering, programming in C, and testing
- Learned KiCad EDA software, gained experience with Visual Studio Code, GitHub, SolidWorks, and microcontrollers
- Gained experience designing analog circuits, reading component datasheets, and using digital interfaces such as SPI
- Focus on safety, reliability, ease of use, clear documentation for future sustainability, meeting budget and time limits

VHDL AND FPGA DEVELOPMENT - Tic Tac Toe (Team Lead And Programmer) – 80+ Hours

APRIL 2025 – MAY 2025

- Led a team of four to meet design requirements, manage deadlines, and integrate hardware and software elements
- Wrote code used in the Tic Tac Toe Game in VHDL for an AMD Artix 7 FPGA using the AMD Vivado Design Suite
- Wrote an IEEE format project report and gave an in-class presentation

ARDUINO, C PROGRAMMING, TECHNICAL DOCUMENTATION - Intelligent Hat (iHat) – 40+ Hours

APRIL 2024 – MAY 2024

- Worked on a team of four to design, construct, and program, and created a formal report, presentation, and video
- Used Arduino programmed in C++ and various peripherals to create a multifunctional wearable device composed of multiple integrated subsystems

SKILLS

- Microsoft Office Suite: Word, Excel, PowerPoint, SharePoint
- Programming : C, C++, Python, MATLAB, VHDL, LabVIEW, GitHub, Visual Studio Code
- Hardware Development and Prototyping: KiCad EDA, Siemens Capital Logic Designer/Mentor Graphics, Arduino, Raspberry Pi, Soldering
- Digital Electronics: Digital Circuit Design, SPI, I2C, UART, Altera Quartus II, Logic Gates, Flip Flops, Shift Registers
- Analog Electronics: Analog Circuit Design, Op Amps, Transistors, Oscilloscopes, Multimeters, Signal Generators, Switched-Mode Power Supplies
- CAD: SolidWorks, Autodesk Inventor
- Manufacturing: 3D Printing, Welding & Cutting Manufacturing Processes, Lathing, CNC